Image Captioning Using CNN - RNN Architecture

```
import torch
from transformers import VisionEncoderDecoderModel, ViTImageProcessor,
AutoTokenizer
from PIL import Image
from transformers.utils import hub

# Set extended download timeout
hub.HUGGINGFACE_HUB_HTTP_TIMEOUT = 60

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# Step 2: Load Pretrained Image Captioning Model
model = VisionEncoderDecoderModel.from_pretrained("nlpconnect/vit-gpt2-image-
captioning")
processor = ViTImageProcessor.from_pretrained("nlpconnect/vit-gpt2-image-
captioning")
tokenizer = AutoTokenizer.from_pretrained("nlpconnect/vit-gpt2-image-captioning")
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
model.eval()

# Step 3: Define Caption Generation Function
def generate_caption(image_path, max_length=30, num_beams=4):
    image = Image.open(image_path).convert("RGB")
    pixel_values = processor(images=image,
return_tensors="pt").pixel_values.to(device)
    output_ids = model.generate(pixel_values, max_length=max_length,
num_beams=num_beams, early_stopping=True)
    caption = tokenizer.decode(output_ids[0], skip_special_tokens=True)
    return caption

# Step 4: Load Image and Generate Caption
if __name__ == "__main__":
    img_path = "download.jpeg" # Replace with any image file
    # Note: Make sure 'download.jpeg' is uploaded to your Colab files
    try:
        print("Caption:", generate_caption(img_path))
    except FileNotFoundError:
        print(f"File '{img_path}' not found. Please upload it to Colab files
before running captioning.")
```

OUTPUT:

```
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.

config.json: 100%
 4.61k/4.61k [00:00<00:00, 263kB/s]
pytorch_model.bin: reconstructing file: 100%
  982MB /  982MB, 49.1MB/s
pytorch_model.bin: downloading bytes:
  910MB, 48.8MB/s
Loading weights: 100%
 445/445 [00:00<00:00, 6902.99it/s]
model.safetensors: reconstructing file:   0%
  0.00B /  982MB
model.safetensors: downloading bytes:
  270MB, 22.9MB/s
[transformers] VisionEncoderDecoderModel LOAD REPORT from:
nlpconnect/vit-gpt2-image-captioning
Key                                                     | Status
|  |
--------------------------------------------------------+-----------
-+--+-
decoder.transformer.h.{0...11}.crossattention.masked_bias | UNEXPECTED
|  |
decoder.transformer.h.{0...11}.attn.masked_bias           | UNEXPECTED
|  |

Notes:
- UNEXPECTED:    can be ignored when loading from different
task/architecture; not ok if you expect identical arch.

preprocessor_config.json: 100%
```

```
 228/228 [00:00<00:00, 21.7kB/s]
tokenizer_config.json: 100%
 241/241 [00:00<00:00, 21.2kB/s]
vocab.json: 100%
 798k/798k [00:00<00:00, 7.05MB/s]
merges.txt: 100%
 456k/456k [00:00<00:00, 5.83MB/s]
special_tokens_map.json: 100%
 120/120 [00:00<00:00, 2.27kB/s]
tokenizer.json: 100%
 1.36M/1.36M [00:00<00:00, 10.2MB/s]
File 'download.jpeg' not found. Please upload it to Colab files before
running captioning.
```